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Industry 4.0/IIoT Platforms for manufacturing systems — A systematic review contrasting the scientific and the industrial side

Holger Eichelberger [a],*, Christian Sauer [a], Amir Shayan Ahmadian [b], Christian Kröher [a]

[a] Software Systems Engineering, University of Hildesheim, Hildesheim, 31141, Germany
[b] Software Engineering, University of Koblenz, Koblenz, 56070, Germany

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ABSTRACT

Context: IIoT, Industry 4.0 or CPPS software platforms are cornerstones of smart manufacturing production systems. Such platforms integrate machines, IIoT and edge devices, realize distributed (management) functionality and provide the basis for user-defined IIoT applications. Individual instances in research and industrial practice do share commonalities while they also differ significantly.
Objective: A detailed overview of the platform landscape is fundamental for innovative research. However, actual surveys and literature reviews concentrate on specific aspects and usually focus only on the research works, neglecting specific aspects of the industrial use of IIoT platforms. We aim at a systematic overview of the functionalities and approaches of scientific and industrial IIoT platforms along 16 analysis dimensions and thereby exposing gaps between the focuses of research on IIoT platforms and actual industrial IIoT platforms in use. By doing so we are able to highlight future areas of interest to research as well as indicating potentially over-researched areas which are of less interest in actual industrial IIoT platforms.
Method: We combine a systematic literature review of scientific IIoT platform research with a systematic analysis of industrial IIoT platforms.
Results: We start off with 1620 research papers plus 70 from snowballing that we systematically filter down to 36 papers (plus 11 added by a SLR update) providing sufficient information for a data extraction, which we analyze along 16 topics to extract actual capabilities and differences of relevant platform approaches. In a second step, we contrast these results with an analysis of 21 industrial platforms.
Conclusion: Similar approaches, differences and topics for future are exhibited. In comparison with 21 industrial platforms along the same analysis topics, we distill various commonalities, differences, trends and gaps.

Contents

1. Introduction 2
2. Related work 3
3. Background 3
4. Methodology 4
 4.1. Search for literature 4
 4.2. Review and filtering of the search results 5
 4.3. Detailed filtering results 6
5. The scientific side: SLR results 6
 5.1. Overview of included papers and approaches 6
  5.1.1. Approach denomination (T1a) 6
  5.1.2. Context (T1b) 6
  5.1.3. Domain (T1c) 6
  5.1.4. Validation methods (T1d) 6
  5.1.5. Access to a realization (T2) 7

* Corresponding author.
E-mail address: eichelberger@sse.uni-hildesheim.de (H. Eichelberger).

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5.1.6. Approaches ... 8
5.2. Extracted results ... 9
5.2.1. Protocols (T3) ... 9
5.2.2. Edge support (T4) ... 9
5.2.3. IIoT devices (T5) ... 9
5.2.4. Security (T6) ... 9
5.2.5. Data protection (T7) ... 10
5.2.6. Cloud support (T8) ... 10
5.2.7. Scalability (T9) ... 10
5.2.8. Digital twins (T10) ... 10
5.2.9. Data management and data analysis (T11) ... 10
5.2.10. AI support (T12) ... 11
5.2.11. Openness and extensibility (T13) ... 11
5.2.12. Systematic configurability (T14) ... 12
5.2.13. Ecosystem support (T15) ... 12
5.2.14. Other technical abilities (T16) ... 12
5.3. Summary ... 12
5.4. Outlook on 2024 ... 12
6. The other side: Comparison with 21 industrial platforms ... 13
6.1. Approach and platforms ... 13
6.2. Extracted results ... 14
6.2.1. Protocols (T3) ... 14
6.2.2. Edge support (T4) ... 14
6.2.3. IIoT devices (T5) ... 14
6.2.4. Security (T6) ... 14
6.2.5. Data protection (T7) ... 15
6.2.6. Cloud support (T8) ... 15
6.2.7. Scalability (T9) ... 15
6.2.8. Digital twins (T10) ... 15
6.2.9. Data management and data analysis (T11) ... 15
6.2.10. AI support (T12) ... 15
6.2.11. Openness and extensibility (T13) ... 15
6.2.12. Systematic configurability (T14) ... 15
6.2.13. Ecosystem support (T15) ... 15
6.2.14. Other technical abilities (T16) ... 16
6.3. Summary and discussion ... 16
7. Threats to validity ... 17
7.1. Construct validity ... 17
7.2. Internal validity ... 17
7.3. Conclusion validity ... 17
7.4. External validity ... 18
7.5. Industrial platforms ... 18
8. Conclusion ... 18
CRediT authorship contribution statement ... 18
Declaration of competing interest ... 18
Acknowledgments ... 18
Data availability ... 18
References ... 18

1. Introduction

Industry 4.0, Cyber–Physical Production Systems (CPPS) or Industrial Internet of Things (IIoT) initiatives aim at realizing the vision of intelligent manufacturing.[1] At its core, intelligent manufacturing requires a continuous analysis of product and production data to steer the production or to enable optimizations, e.g., regarding product quality, down time reduction or energy consumption. This requires interlinking of heterogeneous sensors, actuators, production machines, edge/fog devices and IT systems across various forms of communication, e.g., utilizing standards such as OPC UA.[2]

Managing this heterogeneity and providing a secure IT environment for customized applications and data analytics, including Artificial Intelligence (AI), is a core goal of IIoT platforms. Besides commercial approaches, such as Siemens MindSphere/Insights Hub[3] or PTC ThingWorx,[4] also several scientific approaches are published, e.g., SensorHUB [1], IIoT Bazar [2], AMCoT [3] or MEIX [4]. Since 2013, which can be viewed as the birth year of the Industry 4.0 concept [5], a diverse landscape of software platforms has emerged, each with own capabilities. On the one side, this diversity is an obstacle for researchers in identifying gaps in relevant scientific work. On the other side, it is difficult for companies to make decisions for adopting certain platforms. According to an industrial survey on the future of intelligent manufacturing involving 75 European companies in 2021 [6], only 33% utilize an IIoT platform and only 20% plan for adopting one in the next years. This survey also revealed user issues like licensing or potential vendor lock-ins as well as disregarded demands, e.g., data protection, on-premise installation, or interacting platform ecosystems to mitigate vendor lock-ins.

[1] As IIoT aims at Industrial Internet-of-Things (IoT), manufacturing can be seen as a specific IIoT application. We will use IIoT, Industry 4.0 and CPPS as synonyms.

[2] https://opcfoundation.org/about/opc-technologies/opc-ua/.

[3] https://plm.sw.siemens.com/de-DE/insights-hub/.

[4] https://www.ptc.com/de/products/thingworx.

In this article, we characterize the IIoT software platform landscape from two points of view, the scientific and the industrial side. We analyze the scientific side in the form of a Systematic Literature Review (SLR) based on 36 identified from 1690 candidates (plus 11 added by a SLR update) and contrast the findings with a systematic review of 21 industrial IIoT platforms [7]. We identify commonalities, differences as well as gaps in the current platform landscape. Although several surveys, reviews or SLRs on IIoT platforms are published, e.g., West et al. [8] on ecosystems or Alcacer et al. [9] on technologies, we are not aware of a systematic overview analyzing the scientific and industrial side of the IIoT platform landscape in a similar form addressing the research questions below:

RQ1 *Which IIoT platforms are proposed by research?* We want to complete our understanding of the IIoT platform landscape by identifying approaches and solutions from the research community.

RQ2 *Which application domain(s) and context(s) do the proposed IIoT platforms support?* Domain(s) and contexts may indicate where a particular need for platform solutions may exist.

RQ3 *Which capabilities do the proposed IIoT platforms provide?* We aim at characterizing the capabilities of research IIoT platforms based on 14 analysis topics from [7], which we interactively identified as core features of an AI-enabled IIoT platform and used for collecting requirements [10], designing an architecture and realizing a platform [11,12].

RQ4 *Which differences and commonalities between industrial IIoT platforms and those proposed in research exist?* We also want to reveal overlaps and differences between research and industrial platforms by comparing the results for RQ3 with those from [7] along the same analysis topics. We are convinced that both areas can benefit from the results regarding their future activities in the form that by highlighting the differences we are able to indicate future areas of interest to research as well as identifying potentially over-researched areas which are of less interest in actual industrial IIoT platforms in use.

In addition, we are also interested in the development of the field, e.g., publications over time, as well as addressed evaluation topics, e.g., whether runtime, cost or performance aspects are considered.

Structure: Section 2 discusses published IIoT platform overviews and justifies the need for a SLR. Section 3 introduces the terminology for the analysis. Section 4 outlines our instance of Kitchenham's [13] SLR research method. Section 5 presents the results (and answers RQ1–RQ3). For RQ4, we contrast in Section 6 our results on industrial platforms from [7] with Section 5. Section 7 outlines threats to validity and in Section 8, we conclude.

2. Related work

Several overviews of IIoT platforms have been published, as also noted, e.g., in [14]. We now provide a brief discussion on platform-related overview works, including surveys, (systematic) literature reviews, comparisons and, where applicable, selected extensive related work sections.

Table 1 summarizes published scientific overviews. The works aim at various topics, e.g., edge/fog computing, security, architecture, underlying technologies, data processing or Manufacturing Execution Systems (MES). Among those, seven publications provide an analysis of platform approaches, while the remaining refer to platforms, platform aspects or the notion of platforms, without analyzing them. Only three works in Table 1 apply the SLR research method, covering a similar or earlier range of publication years, a lower number of analyzed dimensions, but a higher number of analyzed papers, however with a differing, particular focus on Big Data [21] or usable platform technologies [23,26,27]. [20] is the only paper we are aware of, which contrasts the scientific and industrial view, while focusing on platform evolution.

Table 1
Related surveys and reviews by years (types: c = comparison, y = survey, l = literature review, s = systematic literature review, r = related work).

Question					SLR		
Ref.	Year	Type	Focus	Platforms	Dims.	Papers	Period
[15]	2017	c	MES	7	(5)	–	–
[16]	2017	r	Fog	9	6	–	–
[17]	2018	l	Security	8	12	–	–
[8]	2018	l	Ecosystems	10	10	–	–
[18]	2018	y	Security	–	7	28	–
[19]	2019	x	Architecture	5	11	–	–
[9]	2019	y	Technologies	–	8	–	–
[7]	**2020**	y	Commercial platforms	21	18	–	-2020
[20]	2020	l	Evolution	–	4	–	2007–2020
[21]	2020	s	Big Data	–	4	128	2008–2017
[14]	2020	r	MES	–	–	5	–
[22]	2021	l	MES	3	7	7	2015–2020
[23]	2021	s	Technologies	–	10	186	2015–2019
[24]	2021	r	Architecture	–	8	18	–
[25]	2023	r	Agent Platforms	13	5	–	–
[26]	2024	l	Cloud-based manufacturing	37	2	–	2010–2023
[27]	2024	s	Platform frameworks	23	1	3146	–

There are scientific overviews on industrial platforms, e.g., [7,19, 28–30] or web pages as listed in [7] including summaries or market analyses.[5] While [19,28,29] discuss comparably few topics/dimensions, the IoT study in [30] covers a broad range of platforms and dimensions. However, in comparison to our overview of 21 industrial IIoT platforms [7], there are common analysis dimensions, but also differences in topics (e.g., we consider AI, edge computing or systematic configurability) and approach (direct interaction with vendors in [30], analysis of publicly available vendor material in [7]).

We conclude that there is currently no systematic overview of the functional capabilities of IIoT platforms contrasting scientific and industrial approaches. Although there are overlaps in the analysis topics, none of the previously published works targets a similar breadth as presented in this article. This justifies a SLR, a prerequisite of the SLR methodology in [13].

3. Background

In literature, fundamental terms for this article are used with similar, identical or differing semantics. We introduce now relevant key terms.

- **Industrial Internet of Things** (IIoT), **Industry 4.0** or **Cyber–Physical Production Systems** (CPPS) are complex, distributed, hardware–software systems operating or optimizing industrial production processes. Although there are semantic differences among these terms, as detailed, e.g., in Lesch et al. [31], we use them as synonyms.
- A **software platform** [12] is a set of coherently integrated software components for a certain purpose. Usually, a platform enables development and execution of user applications based on platform functionality. A platform may be cloud-based or run on premise.
- An **Industry 4.0 or CPPS platform** [12] is a specialized software platform for manufacturing, e.g., integrating with MES or ERP.
- An **ecosystem** encompasses organisms, roles and the physical environment they interact with. A **technical or platform ecosystem** is often based on a (software) platform and shall be open and extensible.

[5] https://www.gartner.com/reviews/market/industrial-iot-platforms.

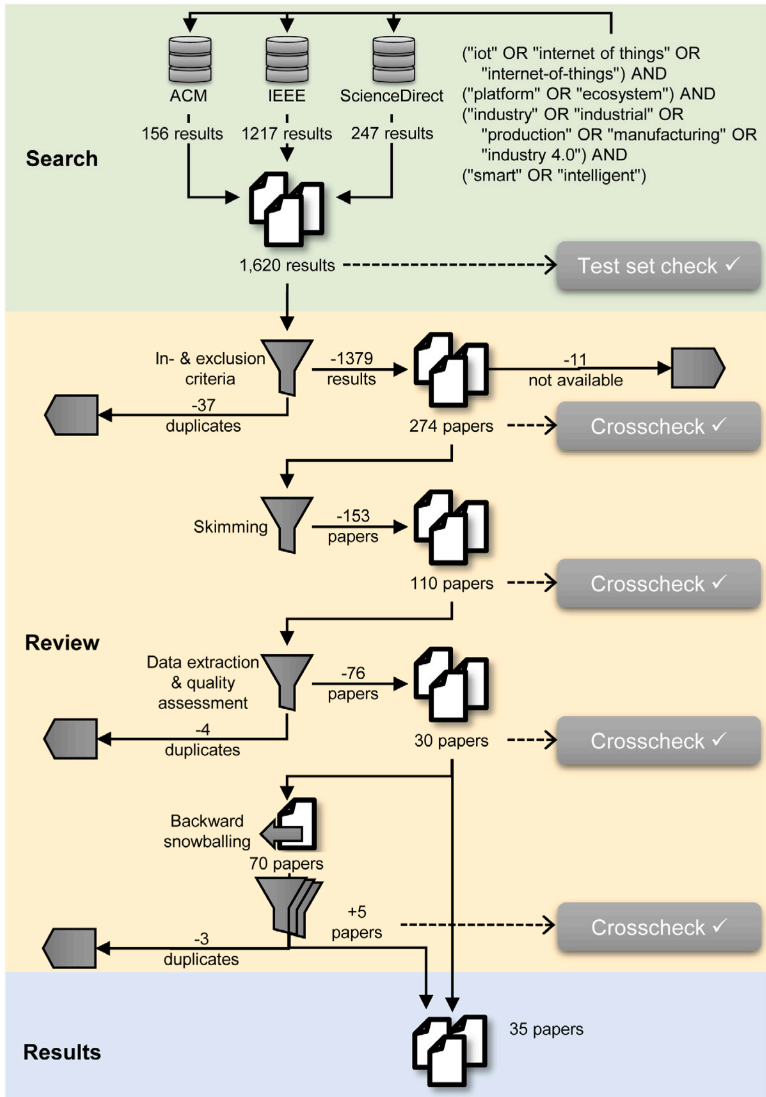

Fig. 1. SLR filtering process.

- A **protocol** defines the language used by entities to interact. Relevant examples are Profinet on field level, the Message Queuing Telemetry Transport (MQTT) or information-model-based approaches like OPC UA or the upcoming IEC 63278-1 ED1 Asset Administration Shells (AAS). Sometimes, also HTTP, REST or JSON are called 'protocol' [7].

4. Methodology

For the SLR we follow the guidelines by Kitchenham and Charters [13]. We also build on our IIoT platform overview [7] for a wider comparison between industrial and research approaches. To answer the research questions from Section 1, we executed the process shown in Fig. 1. It consists of three main activities corresponding to the stages of [13]. We structure this section along the main activities, i.e., *search* for literature (Section 4.1), *review/filtering* (Section 4.2), and abstract *results* (Section 4.3).

4.1. Search for literature

We selected the ACM Digital Library, IEEE Xplore, and ScienceDirect as sources. For the search string, we defined the following areas of interest for which we derived keywords and synonyms:

- *Internet of Things (IoT)* represents the conceptual area of interest. Keywords: "iot", "internet of things", "internet-of-things"
- *Platforms* represents the engineering area of interest focusing on the technological outcome. Keywords: "platform", "ecosystem"
- *Industry* as the area of application we are interested in. Keywords: "industry", "industry 4.0", "industrial", "production", "manufacturing"
- *Intelligence* is an area of interest narrowing the scope to intelligent production as in [7], i.e., platforms integrating Artificial Intelligence (AI) or Machine Learning (ML). Generic keywords: "smart", "intelligent"

The final search string results from disjunctions of keywords per area and the conjunction per area as shown in the upper right part of Fig. 1. We applied this search string to each database using their respective syntax and configuration options to target the title, the abstract, and the keywords and stored the results on January, 8, 2023. Further, we limit the earliest publication year to 2014, i.e., one year after the invention of the Industry 4.0 concept so that first research work could be published. The upper left part of Fig. 1 presents the respective search hits, with a clear majority (76%) originating from IEEE, possibly as the overall topic is more attractive from the engineering rather than the computer science perspective. The coverage of IEEE on the specific topic of engineering records is broader [32], with 70% engineering comparing to 22% computer science topics [33]. At this point, we consider the search hits as "results" rather than actual "papers", as they may include poster summaries, tables of contents, or other irrelevant items.

A *test set check* to ensure the appropriateness of our search string and to guarantee the inclusion of papers in the search results is indicated in Fig. 1. The papers of the test set describe IIoT platforms in the scope of this work and, hence, are (by 100%) included in the results of the search activity.

4.2. Review and filtering of the search results

The review and filtering activity comprises three stages and a backward snowballing. The reviewing process was defined upfront and documented in a guidelines document.[6] We summarize the most relevant details below.

For each of the filtering stages, we distributed the findings on the papers equally between the authors of this paper. Besides processing the assigned papers, each author reviewed the findings by another author as crosscheck. At the end of each stage, all authors discussed their findings and crosscheck results to find consensus in case of discrepancies.

The first filtering stage applies **inclusion and exclusion criteria**, i.e., a result entry must satisfy all inclusion (I), but no exclusion (E) criteria:

- I1 *Peer-reviewed*, like typical workshop, conference, or journal papers
- I2 *Platform as main contribution*, like the introduction of a new one
- I3 *Clear relation to IIoT*, Industry 4.0, CPPS, smart factories, smart manufacturing or smart production

- E1 *Not peer-reviewed or early results*, like keynote, poster, vision, (early) idea, or challenge papers as well as entire books or proceedings
- E2 *Less than six pages*, which does not allow for a comprehensive description of an IIoT platform (we finally exclude many papers below 7 pages)
- E3 *Duplicates*, which either are the same paper (keep only one copy) or describe the same IIoT platform (keep the most complete version).
- E4 *Language*, i.e., the paper was not written in or translated to English
- E5 *Non-available*, i.e., neither the original database for the result nor an additional online search provided a (digital) copy of the paper
- E6 *Published before 2014*, one year after the birth of "Industry 4.0" [5].

In this stage, these (usual) criteria target the bibliographic information, title, abstract, and keywords. This excludes the majority of the initial search results.

In some cases, in particular for I2, I3, and E3, early decisions cannot be made easily. Thus, in case of doubts, we accepted publications to the next stage where a better informed decision could be made.

The second filtering stage is a **skimming** of the papers. This extends the filtering to the entire content and requires access to the papers. From the 273 input papers to this stage, we were able to obtain 262 full texts. Nine papers were not accessible to us (E5) due to licensing issues and two papers were not published in English and excluded according to E4. The focus is on scanning a paper for relevant keywords, phrases, sections, figures or tables. To allow for comparable criteria among the reviewers, we established an early operationalization of platform characteristics (I2) and the IIoT application domain (I3) in terms of the following skimming questions:

- *Does the paper discuss a platform (as introduced in Section 3)?*
- *Is the paper about an IIoT platform for manufacturing or production?*
- *Does the paper discuss an approach to or a realization of a platform?*
- *Does the paper provide details about a platform's capabilities?*
- *Does the paper discuss challenges or problems in developing platforms (e.g., regarding design, realization, etc.)?*

If there are clear indications that in particular the first four questions cannot be answered positively, the paper shall be excluded. This is also the case, if (most of) these questions lead to a negative answer while the last (counter-)question indicates a challenges or position paper.

The last filtering stage consists of **data extraction and quality assessment**. For covering RQ1–RQ3, we devised two general topics (T1–T2), 13 IIoT specific topics (T3–T15) and one topic to collect platform capabilities that are not covered by T3–T15. There are no major intended or expected interdependencies between the topics T3–T15, i.e., we do not expect to have IIoT support if there is also Edge support or that the same protocols are used for global, Edge or IIoT communication. However, some groups of topics are of course related, such as technical topics, e,g, T5 IIoT devices and T11 Data management & analysis or usage topics e.g., T13 Openness & Extensibility and T15 Ecosystem support. The topics are extracted individually and could be implemented/realized by approaches mostly independent, leading to a high number of possible individual combinations of implementations/realizations of topics within a platform. Table 2 summarizes all topics. We identified T3–T15 in interactive vision discussions with research and industry partners in the IIP-Ecosphere project as high-level demands for an AI-enabled IIoT platform. We used these topics for an analysis of industrial platforms in [7], a requirements collection [10] and a platform architecture/realization [11,12]. To answer RQ1–RQ3, we use the same topics, some adjusted to better match research papers, e.g., we removed market and vendor aspects and included scientific evaluation. In summary, we base our analysis on the topics in Table 2, sub-structured into 39 detailing aspects[6]. Hence, each author carefully read the assigned papers to extract and record the information for T1–T15 in a shared data extraction table.

Further, authors also answer the following quality assessment questions:

- *QA1: Is the aim of the research explained?* Valid answers are *implicit* (via explanatory text), *explicit* (via research questions) and *no*.
- *QA2: Is the approach sufficiently explained?* Here we look for a *textual* or a *graphical* overview, a description of the *elements* of the approach or their *interactions*, the *research methodology* or the *evaluation*. The presence of at least three of the indicators leads to a positive answer.
- *QA3: Are the results validated?* (*yes* or *no*)
- *QA4: Is the context of the research stated?* (*yes* or *no*)
- *QA5: Are the research findings stated?* (*yes* or *no*)
- *QA6: Is "the platform" sufficiently described?* We count the presence of data for T3–T15, at least six topics leading to a positive answer.

A paper is finally included, if there are at least three positive answers for *QA1 to QA5* and *QA6* is answered positively. All extracted information as well as the quality assessment are subject to a crosschecking akin to the skimming stage. Although the quality criteria are easy to check, there were borderline cases, e.g., QA6 is passed, but the extracted information is so generic that a characterization of the platform is difficult. In such cases, we made a majority decision in the crosschecking discussion.

A total of 30 papers passed all filtering stages and, hence, serves as basis for **backward snowballing**. Each author considers an assigned subset of the included papers to identify additional relevant work through skimming a paper's references for the title. We processed relevant papers through all previous review stages (details omitted in Fig. 1). In summary, we identified 70 potentially relevant papers through snowballing and after all filtering stages included 5 further papers, yielding a total of 35 papers.

[6] SLR guidelines, extracted results: https://doi.org/10.5281/zenodo.14040398.

Table 2
Data extraction topics (top-level topics only).

Topic		Platform aspects
Id	Concern	
T1	Overview	General information: name, context, domain, evaluation
T2	Accessibility	Download URL and license
T3	Protocols	Supported or used protocols
T4	Edge support	Integration of edge devices, their communication, etc.
T5	IIoT devices	Integration of general IIoT devices, their management, etc.
T6	Security	Supported or used security standards and techniques
T7	Data protection	Processing of (personal) data and its protection
T8	Cloud support	Usage of cloud technology
T9	Scalability	Employed scalability techniques
T10	Digital twins	Supported or used modeling approach and purpose
T11	Data management & analysis	Data processing capabilities, like collection, storage, analysis, etc.
T12	AI support	Supported or used AI/ML algorithms and their integration
T13	Openness & Extensibility	Plug-in/app store for extension and integration of custom code
T14	Systematic configurability	Customization for individual application
T15	Ecosystem support	Connectivity to other platforms
T16	Other capabilities	Features not belonging to T3–T15

4.3. Detailed filtering results

Based on 1620 search hits and 70 papers from snowballing, we considered 1690 candidates in total.

We excluded 83% in the first filtering stage due to partially overlapping reasons: duplicates (E3, 38 papers, 2%), not peer-reviewed (I1, 49 papers, 3%) or short papers (E2, 443 papers, 26%). We further excluded seven snowballing papers published before 2014 (E6). Further exclusions stem from application domain filtering (I3): energy (11% of all excluded papers), agriculture (10%), medicine (6%), smart cities (5%), mobility (3%), or environment (3%). Regarding the agreements, we were unsure about 37 and disagreed on 43 papers (Fleiss' Kappa $\kappa_F \approx 0,85$).

During the second stage, we did not consider 11 papers (around 4% in this stage) as two papers were in Korean or Turkish (E4), respectively, and the remaining were not available to us (E5). Although we searched them on the web and contacted the authors, we could not obtain all identified candidates.

In the skimming stage we excluded 153 of 262 papers for partially overlapping reasons: 13% with still wrong domain, 32% on problems/challenges, 25% on a specific problem rather than a platform, 31% overviews of multiple platforms and 39% without sufficient details. We also found one duplicate (E3) of an approach that was discussed by a more detailed inclusion candidate. In this stage, we had 40 disagreements and 12 unsure ratings ($\kappa_F \approx 0,86$).

The fourth stage started with 109 papers. We excluded 74 papers, 59 as they were not sufficiently described (QA6) and one duplicate. Only 14 papers required an individual decision and led to an exclusion even if the quality assessment indicated inclusion. The reasons were the notion of platform (5), the domain (4), the level of detail (3) as well as duplication (2). A threshold-based quality assessment may not be perfect, but it delivered valuable indications and correct decisions for more than 90% of the papers. In this stage, we had 12 disagreements and six unsure votes ($\kappa_F \approx 0,79$) and we finally included 35 papers. It is important to mention that the authors of [34] present two platforms in sufficient details. Thus, we differentiate from now on between 35 included papers and 36 included platforms/approaches.

Fig. 2 illustrates the distribution of the publication years of all considered papers, classified into exclusions in the first three filtering stages as well as exclusion or inclusion in the final stage. Fig. 2 also indicates snowballing papers, which were published before 2014 and excluded due to E6. The topic of IIoT platforms seems to occur around 2013/2014 with first relevant papers for our SLR published in 2015. Since then, the overall topic became more popular with a certain co-occurrence of the relevant works for this SLR.

5. The scientific side: SLR results

We now focus on the results of our SLR. In Section 5.1, we provide an overview of the included approaches. In Section 5.2, we detail the results for T3–T16, in Section 5.3 we discuss the presented results and in Section 5.4 we provide a systematic outlook on publications from 2024.

5.1. Overview of included papers and approaches

Beside the results for the generic topics Overview (T1) and Accessibility (T2) we present the overview of approaches in Section 5.1.6. Table 3 summarizes all included approaches with graphical indicators for T3–T15 (T16 is optional) illustrating whether we extracted answers at all and, if a topic is sub-structured, the amount of extracted sub-answers.

5.1.1. Approach denomination (T1a)

20 of the included papers (57%) refer to their platform by a name, in 17 papers the name is abbreviated by an acronym (as listed in Table 3). In two cases, a statement on the characteristics of the approach is used instead, e.g., "a message-based modular gateway platform" in [16].

5.1.2. Context (T1b)

As context, we extracted the author affiliations and categorized them into exclusively 'academic', solely 'industrial' or 'mixed'. For 74% of the included papers, the context is 'academic'. Seven papers (20%) are of mixed authorship and only two papers originate exclusively from industrial context.

5.1.3. Domain (T1c)

For the domain we analyze which terms and combinations of synonyms for Industry 4.0/CPPS are used by the authors. The most popular terms are Industry 4.0 (19 times), manufacturing (21) and IoT (20). However, the more recent terms Cyber–Physical Systems (CPS, 9) and CPPS (3) occur less frequently. As typically multiple of those terms are used, we were also interested in popular combinations. The most frequent combinations are Industry 4.0 + IoT (13) as well as (smart) production, Industry 4.0 or IoT + (smart) manufacturing (10 or 11 times each). Combinations with CPPS were used less frequently (1–2 times, e.g., CPPS + IoT or CPPS + IIoT).

5.1.4. Validation methods (T1d)

We analyzed the papers for the applied validation method based on the characterization of Shaw [35]. All 35 included papers applied at least one form of persuasion as indicated in Table 3. However, the frequency of applied validation methods differs as depicted in Fig. 3, color coded along the five validation types detailed below.

- The *persuasion aspect* of a paper is about convincing the reader of the soundness of the approach. This could be done by technique (e.g., walking through an example workflow; 3 papers, 8%), design (e.g., by architectural diagrams; 32 papers, 89%) or example (e.g., an implementation or a proof-of-concept; 24 papers, 67%). 23 papers employ two methods of which the majority use persuasion by design and by example. None of the reviewed papers use all three methods.

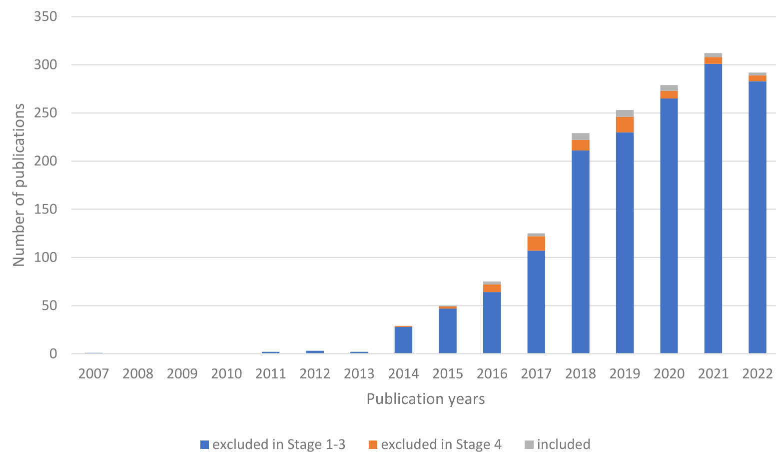

Fig. 2. Distribution of publication years of considered, excluded and included papers.

Table 3
Included approaches by year/author; context (Academic, Industry, Mixed), questions as ○ ◔ ◑ ◕ ● degree of (sub-) answers; evaluation based on [35] as persuation, implementation, system, experience, evaluation, analysis.

Ref.	Year	Name T1a	Ctx. T1b	Acc. T2a	Eval. T1d	Proto. T3	Edge T4	Devices T5	Security T6	Data prot. T7	Cloud T8	Scalab. T9	Twins T10	Data T11	AI T12	Open T13	Conf. T14	Eco T15
[1]	2015	SensorHUB	A	●	pix	●	●	○	○	○	◑	○	○	◑	○	○	●	◔
[36]	2016	CMFAS	A	○	pe	●	●	○	○	○	●	●	◑	●	○	○	●	◔
[37]	2016	Semantic Factory	A	○	pix	●	●	○	○	○	○	○	○	●	○	◔	○	◕
[38]	2016	-	I	○	pia	●	●	○	●	○	●	●	○	●	●	○	○	●
[39]	2017	-	A	●	pex	●	●	○	●	○	○	○	○	●	○	○	●	◑
[40]	2017	-	A	○	pixea	○	○	○	○	○	●	●	○	●	●	◑	○	◕
[16]	2017	-	A	○	pixea	○	●	◑	●	○	◑	○	○	◑	○	◔	●	●
[41]	2018	-	M	○	pixe	●	●	●	○	○	◑	○	○	●	●	○	○	◔
[42]	2018	ICMSMET	M	○	pixe	●	●	○	●	○	●	●	○	●	●	○	○	◔
[43]	2018	4A1C	A	○	pe	○	○	◕	◑	○	○	○	○	●	○	◔	○	●
[44]	2018	ICP4Life	M	○	pixe	○	○	◕	○	○	◑	●	○	●	●	○	○	◔
[34]	2018	MONSOON	A	○	pixea	●	○	○	◕	○	●	●	○	●	●	◑	○	●
[34]	2018	COMPOSITION	A	○	pixea	●	○	○	●	○	○	○	○	●	●	◕	●	●
[45]	2018	-	A	○	pixe	●	●	●	○	○	●	●	○	●	●	◔	○	◔
[46]	2018	UH4SP	M	○	pixea	●	●	○	●	○	●	●	○	●	○	◕	○	●
[2]	2018	IIoT Bazar	M	○	pixe	●	●	◕	●	○	◑	○	○	○	●	◑	○	◑
[47]	2019	SoSM	A	○	piex	○	○	○	○	○	●	○	◑	●	●	◔	○	●
[48]	2019	BPIIoT	A	○	piexa	●	○	○	○	○	●	○	○	●	●	◔	○	●
[49]	2019	-	A	○	piexa	●	●	◕	◑	○	●	●	○	●	○	○	○	◔
[50]	2019	FAR-EDGE DDA	A	●	pie	○	○	◔	○	○	●	○	○	●	○	◔	●	◕
[51]	2019	-	A	○	pixea	○	○	○	●	●	●	○	○	●	○	○	○	◔
[52]	2019	-	M	○	pixe	●	○	◕	●	●	●	●	○	●	●	◔	○	●
[53]	2019	INDICS	M	○	pse	●	○	◑	●	○	●	○	○	◑	○	◑	○	◔
[54]	2020	Snap4Industry	A	○	pixea	●	●	●	●	●	●	●	○	●	●	◕	○	●
[55]	2020	-	A	○	pixe	●	●	○	●	●	●	●	○	●	●	◔	○	◔
[56]	2020	PROPHESY-PDM	I	○	pix	●	●	◔	○	○	○	○	◑	●	●	◑	○	◕
[57]	2020	cogni-IoT	A	○	pixea	●	●	◑	○	○	●	○	○	●	●	○	●	◔
[58]	2020	-	A	○	pie	●	●	◔	◑	○	●	○	○	●	○	○	○	◑
[59]	2020	-	A	○	piea	●	●	◑	●	○	●	○	○	●	○	○	○	◔
[3]	2021	AMCoT	A	○	pie	●	●	◔	○	○	●	○	○	●	○	◔	●	◔
[60]	2021	-	A	○	pi	●	●	◑	●	●	○	●	○	●	○	◔	●	◑
[61]	2021	-	A	○	pie	●	●	●	○	○	●	○	○	●	●	◔	○	●
[4]	2021	MEIX	A	○	pie	●	●	◑	◑	○	●	○	○	○	○	○	○	◔
[62]	2021	-	A	○	pie	●	●	◕	◑	○	◑	●	○	●	●	○	○	◔
[22]	2022	-	A	○	pixe	●	●	◔	●	○	●	○	○	◑	○	○	○	◔
[63]	2022	-	A	○	pixea	●	●	○	◑	○	●	○	○	◑	●	◑	○	◔

- The *description of the implementation* is discerned between the description of technique (28 papers, 78%), e.g., procedures, and the description of a system (7 papers, 19%).
- The *description of the evaluation* could be a descriptive model (2 papers, 6%), a qualitative model (14 papers, 39%) or an empirical quantitative model (17 papers, 47%). Ten papers present a quantitative performance evaluation, among them eight on execution/response time. Only [36] targets (resource usage) costs. Five papers evaluate prediction qualities and two production aspects, e.g., the product cycle time.
- To provide a *detailed analysis*, authors could employ an analytic formal model, e.g., a derivation (1 paper) and/or an empirical predictive model (to predict the results in a controlled situation; 8 papers, 22%).
- 21 (58%) provide a *description of the experience* in multiple, overlapping forms: qualitative model (11 papers), descriptive model (1 paper), decision criteria (2 papers) or empirical predictive model (12 papers).

5.1.5. Access to a realization (T2)

As illustrated in Table 3, four of the included papers (11%) provide access to a realization, among them [39,50] pointing to github,[7] while the others refer to project or company web pages. The artifacts on github are published under open source licenses (Apache 2.0, Affero GPL).

[7] https://github.com/.

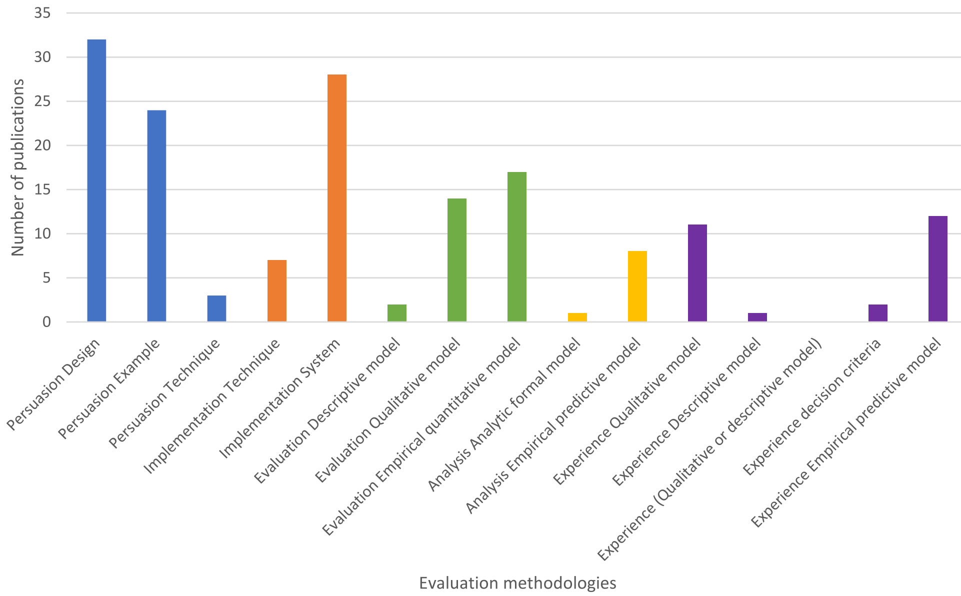

Fig. 3. Evaluation methodologies of the included papers.

5.1.6. Approaches

We now provide an overview of the included papers in sequence of the year of publication. The summaries are based on paper statements.

The publications from 2015 and 2016 mainly follow a cloud-only approach. Lengyel et al. [1] introduce SensorHUB, a cloud-based tool-chain for IoT applications and service development. Similarly, da Silva et al. focus on the development of industrial systems in the cloud through remote monitoring and control. The framework by Chen et al. [36] leverages single-user manufacturing functions to cloud services and scales them in/out based on average service times. Further, Petersen et al. [37] introduce ontological models covering the range from shop floor machines to supply chains.

In 2017, more distributed and contract-based approaches were published. Balint and Truong [39] enable contract-aware IoT dataspace services supporting IoT data flow monitoring based on established contracts. Queiroz et al. present in [40] a CPS, in which agents perform tasks for distributed, collaborative and adaptive process supervision and control. Verba et al. discuss in [16] a messaging-based gateway platform, which allows for clustering gateways and abstraction of peripheral protocol communication.

One year later, first edge/fog-based or cross-platform sharing approaches, integrating platforms as well EU-funded projects occurred. Chen et al. propose in [41] an IoT architecture for distributed edge data fusion. Similarly, Raileanu et al. present a generic architecture [45] for information and data collection, processing and aggregation at the edge. Li et al. introduce in [43] a cross-enterprise knowledge and service sharing framework. In [46], Santos et al. combine cloud computing, IoT, microservices and API gateways to enable data and process integration with third parties. Further, Seitz et al. [2] realized IIoT Bazar, a marketplace for applications with blockchains, fog computing and augmented reality. For integrating intelligent management and remote control, Du et al. propose in [42] a four-layer architecture. From EU projects, the ICP4Life platform discussed by Maleki et al. in [44] is based on a sensing ontology for smart industrial services. Further, the two EU projects MONSOON and COMPOSITION described by Beeks et al. in [34] aim at data intensive processes or platform ecosystem aspects, respectively.

In 2019, publications focused on security, privacy, (manufacturing) integration, configurability and Big Data. Bai et al. [48] aim at lightweight construction of IIoT ecosystems addressing security, trust and on/offline network situations. Also fostering security, Wan et al. introduce in [51] a distributed IIoT architecture with a blockchain-based security and privacy model. Towards integrated platforms, Tao and Qi [47] discuss the combination of modern information technology in manufacturing with service concepts. Dirate et al. integrate in [49] low-cost IoT hardware, smart instructions and live manufacturing process tracking. Kefalakis et al. discuss in [50] how to configure IoT data management using domain-specific languages (DSLs). Yu et al. propose in [52] a CPS architecture for smart, global manufacturing using Big Data analytics. Further, Zhang et al. present in [53] a gateway-based platform, showcased for the energy optimization of central air-conditioning.

In the next year, first open source work occurred and additional views as well as (edge) intelligence were targeted. The open source Snap4Industry platform [54] combines a framework for IoT development with control and supervision aspects. Bosi et al. integrate in [55] operational and business views with device interfaces and communication protocols. Towards more intelligence, the EU project PROPHESY combines in [56] multiple data sources and actionable intelligence. To achieve distributed edge intelligence, Fouklas integrates in [57] fog computing with cognitive IoT.

In 2020, further combinations of security and big data were published. Liu et al. [58] propose a blockchain-based product lifecycle management for data and service sharing. In [59], Yu et al. discuss an IoT architecture for real-time Big Data analytics using existing (open source) technologies.

In 2021, publication topics were middleware, 5G, model-driven development, platform collaboration and quality models. Regarding middleware, John et al. discuss in [60] data transformations between field devices and enterprise applications using EdgeX Foundry. Nkenyereye et al. propose in [4] a oneM2M-based architecture integrating an IoT service layer with multi-access edge computing. Further, Yang et al. present in [62] a cloud approach for condition monitoring using 5G communication and edge computing. The model-driven approach in [60] relies on low-code application development and DSLs for secure predictive maintenance. Regarding collaboration, Hung et al. [3] integrate edge manufacturing services with cloud-based factory-wide and cross-factory manufacturing services. Further, Mocnej et al. introduce in [61] a quality-of-service model for IoT runtime monitoring.

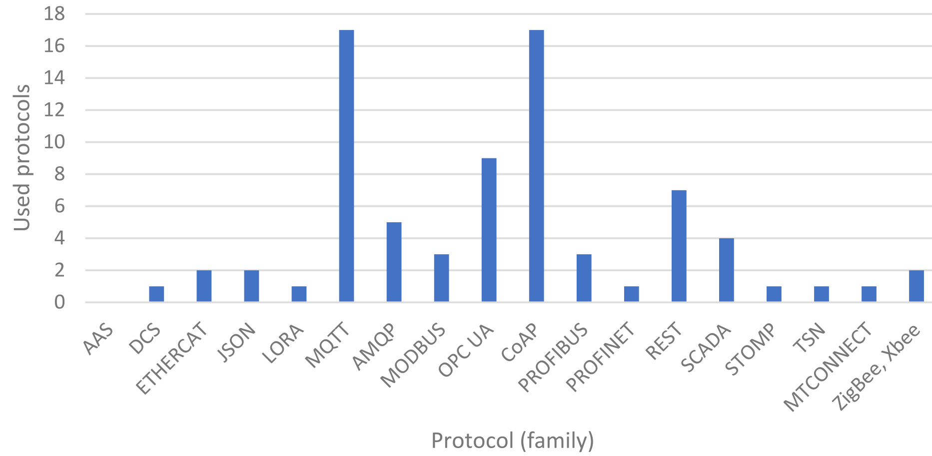

Fig. 4. Stated protocols and information models (ordered by name).

Finally, in 2022, standards, legacy and AI were topics. Mantravadi et al. discuss in [22] a data model based on ISA 95 as well as an ERP integration for operations management in brownfield environments. In [63], Rousopoulou et al. apply a micro-service architecture to data acquisition, knowledge management and predictive maintenance involving continuous AI re-training.

5.2. *Extracted results*

We present now the extracted results for topics T3–T16, each in an own sub-section. Per sub-section, we first focus on the extracted information and in the "*Observations:*" paragraph, we briefly interpret the findings.

5.2.1. *Protocols (T3)*

Fig. 4 illustrates a subset of the extracted protocols and their respective usage frequencies. Most frequently, the authors use application-level protocols like MQTT, Constraint Application Protocol (CoAP) or OPC UA. Field-level protocols as Profinet, Profibus, Modbus or Ethercat are used less frequently. In the figure, we left out common protocols like HTTP/S (11 papers), SSL (1), TCP (4), TLS (1), SFTP (2) or Bluetooth (1). On average, the included approaches employ 2.3 protocols, six papers do not state any protocol, while, e.g., [16] mentions eight or nine different protocols [61].

Observations: MQTT seems to be more popular than its successor Advanced Message Queuing Protocol (AMQP). We can confirm this predominance from practical work with industry in IIP-Ecosphere [11]. For OPC UA, none of the papers detailed the architecture style, i.e., client–server or pub-sub, or whether standardized data formats like OPC UA Companion Specifications are employed. Overall, we expected that around three protocols are used in scientific approaches, so we were surprised about the maximum of 9.

5.2.2. *Edge support (T4)*

For T4, we extracted the support for (explicitly called) "edge" devices in the papers. We focused on the type, role/usage, communication capabilities, forms of memory usage as well as further abilities. 21 platforms (58%) use edge devices, while 22% of the approaches state a *type*: Raspberry Pi (3), UP board [3] (1), SIMATIC IOT20407 [2] (1), Overo AirStorm [45] (1) or Smart IOT 6100 [53] (1). Regarding *role/usage*, 11 approaches utilize edge devices for data processing, six for data collection and four for data fusion. Further, we found production control (3), data storage (2), visualization (1), security enforcement (1), and AI on the edge (1). 10 approaches state edge *communication* protocols, namely OPC UA (6), HTTP/HTTPS (4), MQTT (2), MODBUS (1) or Profibus (1). 9 approaches use edge *memory* for caching or local storage. For 12 platforms, we found *further abilities*, e.g. the use of standards like OCF3[8] or OneM2M,[9] real-time calculations, network function virtualization or smart contracts.

Observations: It appears that industrial edge devices are used infrequently, although the approaches target IIoT. Sometimes generic devices like Raspberry Pi are used, probably as they are cheaper and do not require special knowledge [11]. Except for the devices listed above and one occurrence of a Pi3B+ in [49], the employed devices are not detailed further. Moreover, the utilization of protocols seems to be consistent with Section 5.2.1.

5.2.3. *IIoT devices (T5)*

We focus now on generic IIoT/IoT devices not called "edge". We extracted the forms of communication, device management, deployment or updating capabilities. 25 platforms utilize IIoT devices, eight refer to "sensors", two to "actuators", and some use "PLCs", "robots", RFID reader RC522 [49], triple axis accelerometer MMA8452 [38], temperature sensor DS18B20 [39], thermohygrometer sensor DHT11 [38] or a 6-DOF manipulator [64]. Regarding *communication*, three approaches use MQTT, two OPC UA and two Zigbee. Modbus, Profibus or AMQP were mentioned once. The approaches employ IIoT devices as gateways (5), for storing data in a data lake (1), for blockchains (1) or for real-time communication (1). 13 platforms realize *device management*, e.g., remote monitoring (14) or remote control (9). One platform uses gateway devices, e.g., to compensate network outages. Seven platforms mention on-/offboarding features to introduce/remove devices. Eight platforms discuss the *deployment* of artifacts like software (7), process, AI or decision models (3), or virtualization containers (2). Four platforms support *updates*, for software (3), models (2) or container (1).

Observations: Only a subset of the approaches specify the utilized devices, hiding evaluation details and limiting reproducibility. Any approach uses production machines, e.g., for drilling or welding. In practice, the terms "IIoT device" and "edge" are sometimes used interchangeably, so we were surprised that we could discriminate T4 and T5 without overlaps.

5.2.4. *Security (T6)*

We verify whether security-relevant mechanisms to ensure security requirements such as confidentiality, integrity or availability are used. 23 approaches (64%) introduce at least one *security mechanism*. According to Fig. 5, authorization (11), encryption (8) and authentication (7) are mentioned most frequently. For authorization, the role-based access control mechanism is most frequently stated [42,46,48,51,53–55,59].

[8] https://openconnectivity.org/.
[9] https://www.onem2m.org/.

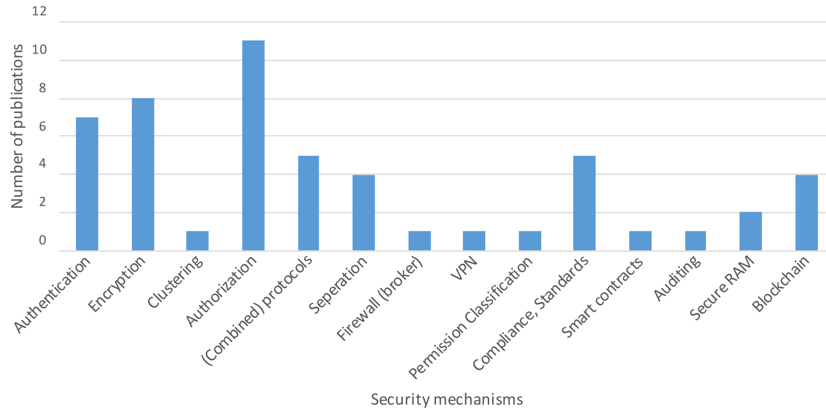

Fig. 5. Security mechanisms used in the included papers.

For encryption, AES-128 [38], Shamir secret sharing scheme [48] and asymmetric encryption [38] are used. The application of secure protocols (5 approaches), separation of concerns (4), adherence to legislation as well as standards (5) and blockchain (4) are stated moderately. TLS, SFTP, HTTPS named as protocols to ensure secure communication and data transferring [22,52,54,60].

Observations: Strong use of the three security mechanisms authorization, encryption and authentication showcases the use of the most known mechanism to ensure security. Mechanisms such as clustering, auditing or smart contracts are not strongly present. Moreover, the authors generally do not reveal details about the utilized mechanisms. For instance, only in [38,48], details about the applied encryption mechanisms are presented, and in fact, these two platforms focus explicitly on security aspects.

5.2.5. Data protection (T7)

We verify whether personal data was processed and the level of data protection supported. Three approaches (8%) offer *personal data* processing. Regarding the *level of data protection*, we extracted the employed data protection mechanisms. 19% approaches provide at least one mechanism to protect personal data: data separation, e.g., by using different physical locations (3), and time limits on data storage (3) are most frequently used. The authors of [54] only state that their research is compliant with the GDPR. 4 approaches state that they use various types of security mechanisms to ensure data protection.

Observations: Personal data processing is not common in the IIoT domain. More approaches (7) offer personal data protection than personal data processing (3), i.e., they offer data protection mechanisms without processing personal data. We observed that mainly security mechanisms are used to ensure data protection aspects. This may indicate that security mechanisms are mechanisms to protect personal data. Moreover, ensuring compliance with the GDPR is rather weak.

5.2.6. Cloud support (T8)

For T8, we extracted, whether cloud technology is used, which type of cloud is applied (public, private, hybrid) and how this relates to the platform, e.g., whether on-premise use is possible or a specific cloud is needed. 30 approaches (83%) rely on *cloud technology*: public cloud (13), private cloud (13), hybrid cloud (9) as well as all three types (8). We were not able to extract the cloud type for 12 approaches. As *particular capabilities*, eight cloud-based approaches also indicate on-premise execution, five approaches that a local and a public cloud is used, and for three approaches, a specific cloud was mentioned: IBM Watson/Bluemix [38], Amazon Web Services [58], Microsoft Azure [61] and IBM cloud [61]. These three approaches also allow for on-premise execution. For 14 cloud-based approaches we were not able to extract further information regarding cloud usage.

Observations: While cloud seems to be a natural choice for modern platforms, the recent trend for (AI on) edge devices in manufacturing may demand on-premise capabilities. Interestingly, around half of 75 surveyed (German) companies in [6] do not want to or cannot use a cloud for several reasons.

5.2.7. Scalability (T9)

If an IIoT platform shall be applied to a shopfloor, scalability matters. 22 of the included approaches (61%) address scalability by the following arguments: It stems from a cloud infrastructure (8), the messaging approach or networking capabilities (8), used components like Apache Cassandra, Apache Hadoop or EdgeX foundry (7), the proposed approach as it is designed for scalability (5), or from the realization as a distributed system (1). Eight approaches employ multiple arguments, one up to three. Most frequent combinations are messaging and cloud (4) as well as messaging and design (2).

Observations: While 39% of the approaches do not address scalability at all, in the identified 22 approaches, scalability is often only mentioned, i.e., scalability is not an evaluation topic in any of the works. Although we just characterize our findings in published approaches, the result is dissatisfying as quality characteristics like scalability, availability or performance are of high importance for industrial production and shopfloor control.

5.2.8. Digital twins (T10)

A current trend in IIoT is the use of digital twins for development, simulation and monitoring of products and machines. Digital twins can be realized through AAS to represent information on an asset in a uniform, interoperable way. For this topic, we extracted the use of digital twins/AAS. Three platforms (8%) *mention* the use of digital twins for modeling assets, usually machines. Two platforms apply digital twins for the *specific purpose* of machine monitoring [47,56] and one platform for machine simulation [36]. None of the approaches mentions the use of AAS.

Observations: These findings are most likely due to the fact that digital twins and AAS are rather recent developments and therefore are not represented prominently in the papers of the reviewed time span.

5.2.9. Data management and data analysis (T11)

Regarding data management/analysis, we looked for capabilities to collect, analyze or store data. Moreover, we extracted statements on distributed processing. 25 platforms (69%) perform *data collection* and 25 platforms (including 17 also doing data collection) employ *data analysis*. 27 platforms (75%) utilize a *data storage*. 16 platforms (44%) distribute the processing or data, two of them into a cloud and [34] between factories. Further, we examined if other forms of data processing are mentioned. 18 platforms (50%) use the data to

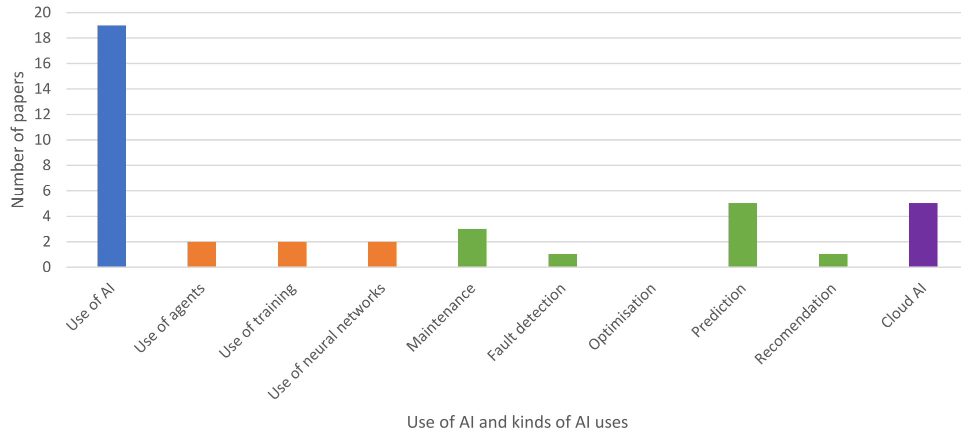

Fig. 6. AI support in the included scientific platforms.

feed AI/ML processes, where AI tasks include decision making (3), predictive maintenance (3) or anomaly detection (1). The ML purposes are feature extraction (2) and model training (1). We also extracted for 6 platforms (17%) that the processing is being executed in *real-time*, out of which three platforms use the term near/soft real-time.

Observations: 50% of the platforms integrate the processing of data with AI/ML, emphasizing their importance among other tasks such as monitoring or visualization. Furthermore, the exchange of data between platforms is only explicitly mentioned for one platform, indicating the relative novelty as well as the potential of ecosystem approaches.

5.2.10. AI support (T12)

The use of AI to monitor, control and optimize existing production processes is a core aim of Industry 4.0. We focused therefore on the use of AI, employed AI techniques, AI capabilities, the purpose of using AI and whether a cloud-based/distributed AI is supported.

With regard to the use of statistical measures and frameworks to establish, for example, the necessary amounts of available data to build and train AI models and suitable thresholds for these AI models, for example, when an AI model should raise an alert, we found only references to either domain experts or the use of best practices. Domain experts are described as evaluating the amount of data and setting thresholds for AI models or using best practices for these tasks.

We analyzed the approaches for the following indicators: The general use of (any form of) AI in a platform and the AI-technique(s) applied, initially focusing on artificial neural networks, any form of training of an AI model or the use of agent-based approaches. We further analyzed the purpose of the AI as one of the following open classifications: maintenance, fault detection, optimization, recommendation or general prediction. Finally, we checked if a platform applied or allowed for cloud-based AI.

We identified 19 approaches (53%) that employ AI as shown in Fig. 6 along the (colored) sub-topics mentioned above. With regard of *AI techniques*, we established that two platforms [41,57] are agent-based and two approaches [40,56] refer to neural networks. Two approaches mention training of neural networks but without details. Regarding the *purpose*, five approaches state prediction, ranging from fault prediction of equipment [45] to predictive maintenance and maintenance scheduling (3). One approach [38] employs rule-based AI to recommend the replacement of IoT devices. Five platforms (14%) use *cloud-based AI*, while none mentioned on-premise use.

In more details, with regard to the AI-methods and techniques we identified the following: In general the use of Artificial Neural Networks (AANs) of varying seizes is the foremost technique being used by the reviewed approaches and in [34] the use of k-Nearest Neighbor, Naive Bayes, Classification and Regression Trees (CART), Random Forests and Support Vector Machines (SVM) is described. The use of Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTM) and Attention-based Networks is described in [56]. The approach by Silva et al. [38] implements a rule-based analytics model. Two approaches are agent-based systems, one with the capability of the agents to negotiate their capabilities [41] and one using cognitive agents with ML classifiers, implementing cooperative incremental learning [57].

To support unsupervised learning we identified the following techniques being used: Majority Voting (MV), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Local Outlier Factor (LOF) and the use of One-Class Support Vector Machines (One-Class SVM), all four being used in [63]. We also identified the use of four clustering performance evaluation metrics to evaluate the AI-models resulting from unsupervised learning: Silhouette Coefficient, Calinski–Harabasz Index, Davies–Bouldin Index and Dunn Index, again all four used in [63].

The development, training and use of AI-models are often supported by tools: We found the use of the following tools in the reviewed approaches: The Protegee Ontology tool for creating and managing Ontologies on which AI-models operate [44], the R4RE system to handle rare events in Predictive Maintenance [56], WEKA to select and evaluate AI-algorithms [40], the JADE-framework to implement agents [40] and the Esper tool for Complex event processing (CEP) [34]. Furthermore, [56,63] employ their own custom build ML-tools. The approaches in [42,47,48,52,54] describe the capability to use cloud-based AI-tools/applications but are not detailing on the specific functionalities that are covered by cloud-based AI tools/applications.

Only some approaches specify the programming language used to implement the AI-functionalities, four approaches mention Java [34,40,44,56] whilst one approach uses Javascript and Python [63].

Observations: Surprisingly none of the included approaches explicitly mentions optimization as a purpose. However, this might be an implicit assumption of the approach.

The low number of approaches mentioning (the training of) neural networks was unexpected given the popularity of that AI technique. Further, it seems that cloud-based AI is yet adopted reluctantly, while on-premise AI is probably the implicit application. The use of Python in only one approach was surprising, as it is recently becoming the dominant language in the development and use of AI-models.

5.2.11. Openness and extensibility (T13)

To cope with different setups, IIoT platforms need to be open and extensible. As this work was motivated by the IIP-Ecosphere project, in which interoperability, use of standards and openness were explicit goals, we extracted whether a (data or app) store, support for app or content developers, or the ability to share or use external data or algorithms is present. Three of the approaches offer a *store* for applications [2] or microservices [46,54]. *Support for developers* is stated by six platforms (17%). In this context, five approaches (14%) mention *interfaces* to customize and/or develop new AI-solutions. The abilities to *share and use external data or algorithms* is described by 19 (53%) of the platforms.

Observations: We found certain industrial features on the scientific side, e.g., active support for developers. The low number of stores may be due to the academic origin. Further, also academic platforms aim for openness, e.g., to allow third parties to contribute. This indicates a trend away from "walled gardens", a frequent characteristic of commercial platforms.

5.2.12. Systematic configurability (T14)

Here we looked for systematic forms of customizing software systems, e.g., Software Product Lines [65], as often a one-size-fits-it-all approach does not satisfy all customers [6]. 15 approaches (42%) mention the following configuration techniques: The use of a configurable frameworks or software components (5), the ability to modify the network configuration of devices or system components (3) or models representing parts of the system (3), in [60] in form of DSLs. Three approaches use configuration files to customize parts of the system and one approach targets customized device installations.

Observations: We expected more cross-fertilization between configuration and IIoT platforms, as production machines are often highly configurable. While the three model-based approaches may indicate a configuration approach, others apply only basic techniques and do not target system consistency. However, it was difficult to extract configuration techniques as this topic is not in the focus of the approaches.

5.2.13. Ecosystem support (T15)

We are interested in capabilities that enable the creation of platform ecosystems, i.e., systems of systems that exchange data or services. Here we focus on whether a platform is (a) "multi-sided" and networks with other platforms, (b) open to third-party content, services, devices or data and (c) based on any reference architecture to ease interoperability. 17 approaches (47%) provide at least one answer. For 15 approaches we found indications for *multi-sided capabilities* and for 16 platforms we identified *openness regarding data or content*. Further, 14 platforms aim at integrating *external services*, 14 platforms support accessing external data and 11 allow for using external devices. Frequently mentioned combinations were data + services (11), devices + services (10) as well as devices + data (10). Regarding *reference architectures*, RAMI 4.0 (DIN 91345)[10] was used in [22,55], oneM2M[9] in [4] and ISA 95 in [22]. Further, as a side action, we extracted architectural statements like service-oriented (6), cloud-based (5), micro-service-based (4), blockchain-based (4), Hadoop-based (4), and event-based (3) architectures.

Observations: We found only three reference architectures in four approaches (10%) targeting an application domain with high adoption of standards. In this context, [34,45,54] defined an own reference architecture.

5.2.14. Other technical abilities (T16)

For T16, we openly collected technical capabilities that are not covered by T3–T15. For 20 platforms we recorded the following capabilities: Three approaches rely on blockchains [2,48,66] to increase security or transparency, while [48] also applies smart data usage contracts. Further abilities are augmented reality [2], no/low code concepts [50], or ontologies [36,37,44]. Moreover, several platforms apply virtualization: Eight approaches utilize Docker, six employ containers without mentioning a specific technology and eight platforms rely on virtual machines, while some combine multiple technologies, e.g., [3,63] Docker containers and virtual machines.

[10] https://www.plattform-i40.de/PI40/Redaktion/DE/Downloads/Publikation/din-spec-rami40.html.

5.3. Summary

From the extracted information, we identified a diverse set of capabilities, which helps gaining a better understanding of the research landscape of Industry 4.0, IIoT and CPPS platforms (RQ1). A high-level summary of extracted topics is given in Table 3. Moreover, Fig. 8 provides a different view on Table 3 comparing scientific and industrial platforms. Many of the identified approaches are cloud-based, some focus more on the IoT side [1], on semantics [37,44], security [48,60], data analysis [40], or platform interconnectivity [43]. Typically, the platforms are distributed, reflecting the nature of the underlying installations. Recent approaches like [3,57,63] include edge/fog computing or AI.

Regarding domains (RQ2) we identified that the term "Industry 4.0" is as frequently used as (smart) manufacturing or IoT, while IIoT, CPS or CPPS are used less often. More than 70% of the papers are of academic authorship, while the remainder is mostly mixed.

The main focus of our extraction was on the capabilities of the proposed platforms (RQ3). Depending on the application field, different *protocols* are used, while recent work mostly apply OPC UA or MQTT. More than half of the approaches address *edge* devices, whereby AI on the edge is a potential gap. More than 70% employ "classical" *devices*, however, often without detailing the employed devices. Concerning *security and data protection*, we observed that mostly known security mechanisms are used. Often, the included approaches do not detail privacy or data protection. Besides better support also for trustful data sharing, a wider uptake of GDPR would be desirable. More than 80% of the approaches are *cloud-based* and about a quarter mentions on-premise computing, while flexible edge–cloud approaches are missing. Although *scalability* is important in the production domain, about 60% of the authors touch the topic while neither arguing scalability in detail nor evaluating it. Only 8% approaches use *digital twins* and none applies AAS, although this is also a trending topic in IIoT. Further, 92% of the approaches perform *data management*, in about 50% to integrate with AI or ML methods. The exchange of data *between platforms* is only mentioned by a single approach, a potential field for future systematic work on interoperability. *AI* was used by more than half of the approaches, whereby only four approaches provide details. Further, the included approaches do not frequently dig into *openness, extensibility or ecosystems*. We also identified potential for *systematic configurability* as 40% only address basic aspects of this topic, usually mentioned without details.

While all papers provide some form of validation according to [35], the amount of approaches decreases the more expressive the validation technique becomes. Most of the papers providing a quantitative evaluation target performance (execution/response time). Future evaluations on runtime aspects can help closing the gap for quantitative cross-evaluations among platforms. However, as only 11% of the papers provide realization artifacts, most of the discussed results can neither be validated nor replicated independently.

5.4. Outlook on 2024

As SLR results may quickly become outdated, we report now on recent publications that we identified by applying the SLR update methodology of Wohlin et al. [67]. However, as discussed in [67], newer publications cannot simply be integrated into completed SLR results. Following the update guidelines, we conducted mid-November 2024 a forward snowballing on the included papers from Section 4.3 to identify candidates published in 2024 and processed them with our SLR filter chain from Section 4.2 including crosschecks and consensus meetings for all obtained results. We discuss first the filtering, provide then a brief summary of the included papers and finally indicate their contributions to our analysis (in particular to RQ1 and RQ3).

For each included paper from Section 4.3, we looked up the citing papers from 2024 in Google Scholar (337 papers) and determined relevant candidates through an analysis of title and abstract akin to stage

1 of our SLR process ($\kappa_F \approx 0,85$). Thereby, we excluded 306 papers based on our inclusion/exclusion criteria from Section 4.2, mostly due to topic mismatches (I2, I3), e.g., blockchain approaches to Industry 4.0 security or for being a literature-based work like a review or a SLR (71 works), due to the language (E4, 18 papers), as they are not yet published (E1, 14 works) or as we identified papers as duplicates (10). Among them, seven papers were not accessible to us due to IPR or license reasons (E5); none of the approached authors provided a copy of their work. For the accessible 28 papers, we excluded 12 papers during skimming ($\kappa_F \approx 0,78$) and analyzed 16 papers in detail. Based on the extracted information and the quality assessment, we further excluded five papers ($\kappa_F = 1$) and report now on 11 relevant papers published in 2024. It is interesting to note that for 2024 we found more relevant papers than for all individual years addressed by our SLR before.

From an overall perspective, the identified papers focus on four topics, namely AI support, digital twins, cloud/edge deployment as well as data security. Regarding *AI support*, Wang et al. propose in [68] a knowledge-driven autonomous manufacturing system using a pre-trained transformer model, while [69,70] aim to support human aspects through AI. Alberti et al. construct in [69] data pipelines for human-centered AI manufacturing systems, and Kotsiopoulous et al. aim at defect detection using explainable AI and human-in-the-loop approaches in [70]. Two platform works specifically target the construction of *digital twins*, namely Yang et al. in [71] focusing on plug-and-play twin functionality and Simoes et al. [72] on composable/generated user interfaces. *Cloud, edge or fog deployment* is addressed by five works: Cuadra et al. employ model-driven techniques in [73], Xiao et al. [74] realize an interactive network condition monitoring system based on edge computing, while Liu et al. in [75] and Xiao et al. in [76] focus on collaborative CNC machine control. Thereby, [76] supports the STEP-NC standard, and Li et al. aim at a digital manufacturing system as a platform-as-a-service providing virtual reality for simulation and quality assurance in [77]. Finally, Tan et al. present a resource collaboration architecture for *data security and sharing* through blockchains in [78].

With regard to our analysis topics, we discuss now noticeable trends, leaving out topics for which the new papers do not provide a contribution:

- We identified a *context (T1b)* shift towards mixed affiliations (increase by 34%) from exclusively academic affiliations (from 74% to 36%).
- The *access to a realization (T2)* slightly increases from 11% to 18%, as [72,73] published artifacts on github under MIT or GPL-3 license.
- Also the support for *edge devices (T4)* increases from 58% to 73%. While only [76] provides details on the employed devices, now a larger share of 18% runs AI tasks on the edge [68,69] for prediction, classification, or training as well as to improve security or data privacy [78].
- In contrast, only 27% of the works explicitly rely on *IIoT devices (T5)*, i.e., a reduction of about 40%.
- 7 publications (64%) in 2024 introduce at least one security mechanism. Our SLR results in Section 5.2.4 with respect to *security (T6)* similarly indicated that 64% of the approaches introduced at least a security mechanism. Authentication, encryption and blockchain are mentioned most frequently in 2024. This is almost comparable to our SLR results where authorization, encryption and authentication were stated most frequently.
- With respect to the *data protection topic (T7)*, only one publication (9%) introduces a data protection mechanism. This is slightly fewer than our SLR results (Section 5.2.5), where 3 approaches (19%) provided at least one mechanism to protect personal data.
- Compared to our SLR, a similar share of 73% approaches relies on *cloud technology (T8)*. However, for the more recent papers, none explicitly states a cloud provider, [69,78] explicitly indicate edge as well as cloud execution and only [77] mentions on-premise execution.
- Statements on *scalability (T9)* made in 36% of the 2024 works are similarly weak and remain unevaluated as in the SLR findings; three papers argue scalability through distributed or cloud processing, one by virtualization and one by data storage technology.
- The use of *Digital Twins (T10)* has significantly increased in the 2024 publications: Five works (45%) describe the use of digital twins, compared to only three publications (8%) in our SLR results. The upcoming standard for describing assets in Industry 4.0 and IIoT, the Asset Administration Shell (AAS), was mentioned by none of the 2024 publications, which is consistent with our SLR results.
- With respect to *data management and analysis (T11)*, only one publication in the 2024 update (9%) mentions data distribution which is significantly less than our SLR results (44%). For data analysis, storage and collection, the results in 2024 are comparable to our SLR results.
- *AI support (T12)* is described in 55% of the 2024 publications, which is consistent with 19 publications (53%) in our SLR results. We found three particular novel aspects: the inclusion of eXplainable AI (XAI) and Human-Centered AI (HCAI) covering the entire AI-Life-cycle and the use of Large Language Models (LLMs). One 2024 publication [74] (9%) employs a statistical approach (three-sigma) to define AI-model thresholds, compared to none in our SLR results.
- Regarding *configurability (T14)*, we identified an increasing number of papers (by 24%) employing model-driven/low-code approaches, a registry [75] or allowing to configure knowledge discovery tasks [69]. However, the overarching aspect of systematic or consistent configurability is still not targeted by any paper.
- In the topic of *other technical abilities (T16)*, the number of container-based approaches raised by 15% in particular employing Kubernetes. Further, the platform in [74] is based on a "comprehensive algorithm module library", which can be seen as a systematic approach to reuse.

The radar chart in Fig. 7 illustrates a comparison between the SLR results and publications in 2024 with respect to the fractions of platforms answering the respective topics T3–T16. It particularly illustrates the reduced support for IIoT devices, a lower number of statements on scalability, data distribution or ecosystems. In contrast, as summarized above, more approaches focus on digital twins, data collection, data analysis or openness.

6. The other side: Comparison with 21 industrial platforms

IIoT software platforms are attractive to both, the scientific and the industrial side. We now present a comparison of these two sides, where applicable following the same structure as in Section 5.

6.1. Approach and platforms

The basis for this comparison is the set of overlapping analysis topics from our overview of industrial platforms [7], where we systematically analyzed the public vendor documentation of 21 platforms. The platforms to be analyzed were thereby determined by two significant sources:

- A *stakeholder analysis* which identified platform candidates for a more detailed analysis and prioritized them. The prioritization was based on revenue and profit figures as well as relevance to the Industry 4.0 approach. We considered platforms with the largest market share, i.e., Siemens MindSphere, PTC ThingWorx, SAP

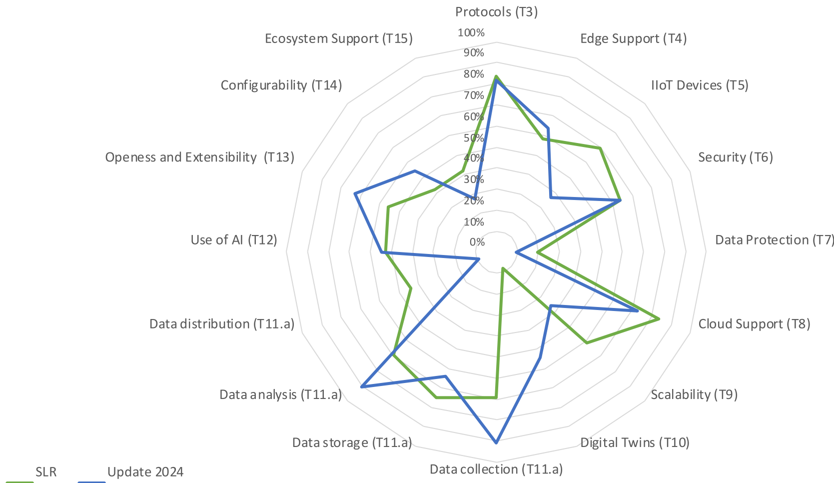

Fig. 7. Comparison of our SLR and the publications in 2024.

Leonardo, Software AG Cumolocity, Bosch IoT Suite, Cisco Kinetic, General Electrics Predix, Microsoft Azure IoT Suite, Amazon AWS IoT, and Oracle Cloud IoT.
- The *discussion of the IIP-Ecosphere project experts* who are particularly interested in developing an Industry 4.0 platform and its architecture. The experts belong to two scientific institutes, and five leading tech companies in Germany. The platform candidates were of special interest to the IIP-Ecosphere project like DeviceInsight Centersight, Weidmüller Industrial Analytics, Harting MICA, Endress + Hauser Netilion, or Adamos.

About half of the industrial platforms originate from Germany and 38% from USA. 90% of platforms are commercial, while two do not provide any information about their license model. A third of the platforms mention the use of open source libraries, while none of the platforms is open source. For none of the platforms, we found information on runtime qualities or performance statements while costs are typically subject to B2B discussions.

6.2. Extracted results

In this section, we discuss the findings from [7] in subsections for T3–T16 including observations that we made. In Section 6.3 we provide a summary.

6.2.1. Protocols (T3)

Many protocols are used in IToT also, as older or legacy machines need to be retrofitted. 80% of the platforms offer a wide variety of protocols, sometimes more than a hundred are mentioned. We identified 37 different protocols, among them HTTP/REST/JSON (15), MQTT (15), MODBUS (10), OPC-UA (10), and AMQP (4).

Observations: We found an overlap with the scientific side, e.g., OPC UA, MODBUS, and MQTT. We see a trend towards (a few) standardized protocols such as OPC-UA and MQTT besides the ubiquitous use of HTTP/REST for API access. Moreover, scientific platforms tend towards a few exemplary protocols, while industrial platforms aim at being more encompassing.

6.2.2. Edge support (T4)

Edge devices are a trend in the industry, which is taken up by 86% of the platforms. No information was provided for the remaining platforms. However, the kind of use varies significantly: 52% support cloud *storage*, while 67% support both, cloud and local storage. Regarding *data processing*, 48% mention data pre-processing, 52% edge analytics and 38% long-term edge data storage. Only 29% allow for execution of *customer-specific applications* on the edge. Likewise, only 33% of mention the execution of AI/ML. Four platforms support Docker containers. Regarding *communication*, 52% offer gateway software and 62% mention MQTT support, while 38% mention *edge–cloud* communication. Finally, regarding *management*, 33% provide a dedicated edge management component and only four platforms provide components to manage software updates, the edge device life cycle or connected IoT devices. GE Predix, AWS IoT, Bosch IoT Suite, PTC ThingWorx, SAP Leonardo, or Siemens MindSphere rely on dedicated edge software. In addition, four platforms mention *digital twins* for the edge, AWS IoT and Cloud IoT allow for semantic modeling of devices.

Observations: Edge devices are also considered by a majority of the scientific approaches. However, more industrial approaches realize data storage or AI on the edge. Further, both sides have oppositional preferences regarding OPC UA and MQTT, where the scientific side prefers OPC UA whilst the industrial side prefers MQTT. In summary, the identified capabilities are rather diverse, both sides lacking standardization for edge computing.

6.2.3. IIoT devices (T5)

95% of the platforms indicate specific support for IIoT devices. Regarding *connections*, 29% specify the technology used or required, e.g., gateways for Bosch IoT or Kinetic. For *device management*, 42% mention on- and off-boarding features, while 33% offer a form of automation like device templates. A similar subset supports remote access or monitoring. 24% support the device life cycle, e.g., backup, restore or rollback. Regarding *deployment*, 38% enable over-the-air updates and 19% support containers, usually based on Docker, for individual functions (AWS IoT), microservices (Centersight) or ML models (Google Cloud IoT Core). 14% rely on specific libraries or operating systems, e.g., FreeRTOS or Yocto Linux.

Observations: In contrast to the scientific side, protocols are not specified explicitly by the industrial platforms. Surprisingly industrial abilities like device management, monitoring, remote access and – less frequently – on- and off-boarding or deployment are relevant to scientific approaches.

6.2.4. Security (T6)

90% of the industrial platforms state that they provide various *mechanisms to ensure security*. Only for two industrial platforms we were not able to identify any type of security mechanism. Further, 90% state that they fully support cryptographic mechanisms. Moreover, 86% provide full support to prevent unauthorized access to network services.

Observations: Only 22% of the scientific platforms support encryption and 30% mention authorization for access management. The support for security on the industrial side is much stronger, which might be due to obligations to ensure quality aspects including security in a competitive market.

6.2.5. Data protection (T7)

76% of the industrial platforms recognize personal data to understand the applicability of the GDPR. Moreover, in case of limited data storage and processing as a main mechanism to realize *data protection* requirements, 81% of the industrial platforms provide full/limited support.

Observations: In contrast to security, the differences for data protection are even larger. Only 8% of the scientific platforms state that they *process personal data*, and further 8% support time limits on data storage. The reason might be the articles stipulated in the GDPR on remedies, liability and penalties. Infringements are subject to administrative fines and, therefore, personal data aspects are more important to industrial vendors.

6.2.6. Cloud support (T8)

95% of the platforms are integrated with a cloud: 19% detail their *cloud integration* as optional, while for most of the remaining platforms cloud services are mandatory. For 24% an *on-premise* installation is possible, while only Adamos allows for combinations. 15% state support for multi-tenancy. Regarding cloud providers we found Microsoft Azure (12), IBM (3), Amazon (3) as well as Huawei, Alibaba, or the platform vendors themselves.

Observations: A similar share of the scientific approaches (83%) do rely on cloud services and a similar fraction (22%) aims for on-premise. The following cloud providers do overlap between the sides: Azure, IBM, Amazon.

6.2.7. Scalability (T9)

81% of the platforms mention scalability: 29% relate scalability to the addition or removal of devices, 38% name abilities for runtime scaling, e.g., load balancing, and 33% base their argumentation on cloud capabilities.

Observations: 61% of the scientific platforms argue scalability, mostly through cloud technology, as the only identified overlap between both sides.

6.2.8. Digital twins (T10)

Four industrial platforms mention digital twins: 12 (57%) employ the *concept of digital twins*, albeit in different forms, e.g. focusing on digital representation of entities/devices or simulation. The development of new devices, device configurations or business processes based on digital twins is supported by eight platforms (38%). Digital device shadows are supported by six platforms (27%). *AAS* is not mentioned by any platform while seven platforms describe similar approaches to a uniform modeling of entities.

Observations: Given the comparably strong representation of digital twins on the industrial side it is remarkable that we found indications in scientific work only in [36,47,56], all aiming at the monitoring of machines. This may be due to the relative novelty of the topic and may take up a broader coverage in future as this already happens, e.g., in the ETFA conference series.

6.2.9. Data management and data analysis (T11)

19 of the industrial platforms (90%) offer functionalities for the collection and storage of data with a majority using cloud services. 57% describe themselves as *real-time* capable for data collection and processing. All platforms provide *data analysis* features, with 38% naming details.

Observations: The availability of data collection correlates with 25 (69%) of the scientific platforms. Further, 27 (75%) of the scientific platforms apply data storage, often based on cloud services (83%). Real-time or near real-time capabilities were only stated by six scientific approaches (17%).

6.2.10. AI support (T12)

15 (71%) of the industrial platforms offer AI support. Regarding *use cases*, we found anomaly detection (33%) and behavior prediction (19%). 52% offer AI programming interfaces and 48% provide AI through the platform's own AI procedures. 14% allow the use of external AI applications.

Observations: Compared with the scientific side we identified a strong focus on AI support for 71% of the industrial and 53% of the scientific platforms. A notable difference is in AI interfaces, which are offered by 52% of the industrial and only 14% of the scientific platforms. This could be due to the fact that the scientific platforms are often prototypical and focus more on evaluation rather than on the provisioning and customization of AI solutions through AI interfaces, as in industrial platforms.

6.2.11. Openness and extensibility (T13)

For this topic, we looked for a data or app store, developer support and approaches to openness and extensibility. 48% offer a *store*, e.g., for services (38%) or applications (38%). Moreover, 17 platforms (81%) provide *developer/customer support*, e.g., tutorials (48%) or platform documentation (48%). With regard to *openness and extensibility*, 67% of the platforms allow the use of external data or external algorithms (62%).

Observations: The low number of stores on the scientific side (8%) can be seen as an effect of the academic context, e.g., as the platforms are not yet mature enough. In contrast, on the industrial side, a store is often a key for monetizing a platform. Similarly, supporting material may be needed to entice developers or customers to develop for and use content from a store. It is worth to mention that both sides allow on a similar level for external data and algorithms, probably an intent to become used or and established.

6.2.12. Systematic configurability (T14)

Akin to the scientific side, we collected mechanisms for configurability rather openly. For 81% we identified the use of one or multiple techniques: 38% allow for customization of *applications*, data analyses, rules, execution schedules or KPIs, e.g., by modeling the IoT environment (ThingWorx), the access to APIs (Adamos) or properties of components or services (B&R mapp Technology, Oracle Cloud IoT). *Pre-configured software packages*, are offered by 42%, which can be further customized in seven platforms. 19% allow for the customization through *visualization elements*, e.g., dashboard widgets. For 29% we found indications that the *platform* can be customized, e.g., through re-branding or the presence/absence of components or services.

Observations: In contrast, 42% of the scientific approaches mention configuration abilities, but both sides do not detail their configuration abilities.

6.2.13. Ecosystem support (T15)

A platform may act as the core of an ecosystem, e.g., interlinking the partners or allowing for the exchange of components and solutions, but may also cause an undesired vendor lock-in [6]. 67% of the industrial platforms can, to varying degrees, be characterized as a *multi-sided platform*. While Amazon AWS IoT, Harting MICA, IBM Watson IoT Suite and Oracle Cloud IoT focus on their own ecosystem, 12 platforms explicitly offer networking with other platforms. 38% state that external platform (cloud) *services* can be integrated, 67% enable integration *third-party* functionality, which, in case of four platforms, are limited to certified partners. Moreover, 43% of the platforms allow the integration of *third-party data*. Further, 38% offer *company networks* for closer collaboration. Regarding *reference architectures*, we found that only PTC ThingWorx refers to RAMI 4.0.

Observations: Both sides indicate at a similar level some form of ecosystem support, usually driven by a different motivation. Further, we found on both sides some (but not many) uses of reference architectures.

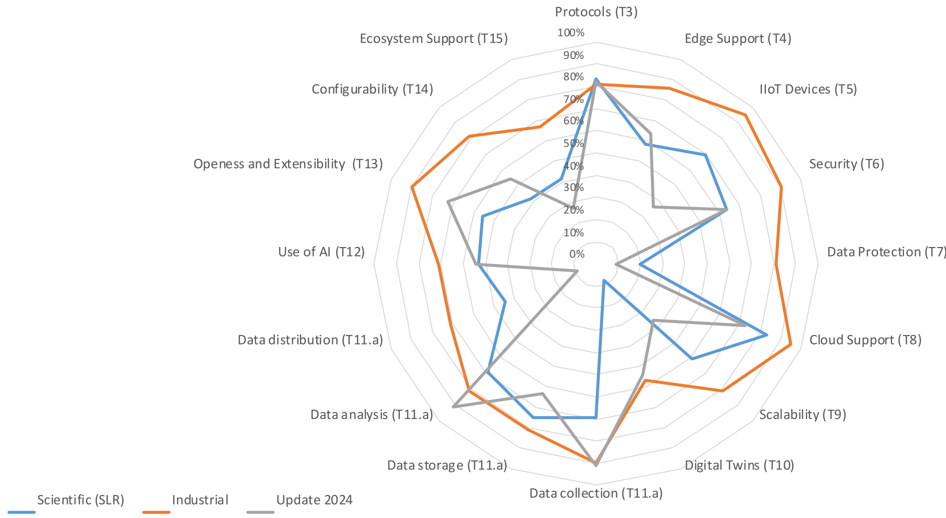

Fig. 8. Comparison of both sides based on answers to analyzed topics. The radar chart further contains the results of the SLR update (Section 5.4) where scientific papers published in 2024 are analyzed and reported.

6.2.14. Other technical abilities (T16)

We found abilities like virtualization techniques (38%) usually based on Docker and, less frequently, Kubernetes, microservice architectures (38%), graphical programming approaches (24%), as well as location-based technologies, speech processing, virtual/augmented reality for maintenance tasks, or device shadows to handle offline states or connectivity issues.

Observations: While there is an overlap between both sides, we also found unique technical abilities on the scientific side, e.g., blockchains or ontologies.

6.3. Summary and discussion

The comparison between the SLR results and our findings for 21 industrial platforms is the basis for answering RQ4. Scientific and industrial IIoT platforms share several trends, show similar gaps and also differ. The two radar charts in Figs. 8 and 9 illustrate that with respect to the fraction of platforms answering the respective topic and sub-topic. The two figures further illustrate a comparison to the results of the SLR update described in Section 5.4, where 11 new relevant papers published in 2024 are analyzed (gray line). For the scientific side, Fig. 8 is a different view on Table 3. For the Data Management and Data Analysis topic (T11) in Fig. 8 we included four aspects of the sub-topic Data Processing Overview (T11.a) rather than one aggregated value. In Fig. 9, we only included a selected number of sub-topics. Below, we will detail the individual topics.

Protocols (T3) is the only topic for which (marginally) more scientific platforms provide information than industrial platforms. However, both sides do not take up more standards like AAS or OPC UA to increase interoperability, while specific approaches for this aspect do exist, e.g., [79]. Overall, OPC UA is provided more in the industrial side.

Industrial platforms provide notably higher support for edge computing (T4) and IIoT devices (T5). Some industrial approaches tend to support only specific device types, which may cause a vendor lock-in. Where such a focus exists, the capabilities of the employed devices often remain unclear. For T4 and T5, some perceived gaps are AI on the edge, run-time management of user-supplied components or fog computing.

While both sides describe how security mechanisms (T6) are supported, industrial platforms demonstrate higher emphasis on security features (such as authorization and cryptography). Further, industrial platforms significantly prioritize data protection (T7), particularly, limited processing is highly supported by the industrial side. We could imagine, that more integrated work on platform security, security by design as well as (issues) complying with regulations like GDPR traded off with openness and ecosystem demands could be an interesting future topic.

Both sides embrace cloud technologies (T8). However, industrial approaches tend to neglect user concerns like fear of vendor lock-ins, on-premise demands, limited network bandwidth, too high latency or missing trust in clouds [6]. Further, clouds are often used to argue scalability (T9), while both sides largely miss explicit demonstration or evaluation of scalability.

Regarding digital twins (T10), industrial platforms demonstrate a more robust commitment, however in the analyzed platforms they do not agree on an interoperable standard. Digital twins in general and AAS as a specific realization are only reluctantly taken up in platform work, although, with advancing format specifications, e.g., as provided by the OPC UA Companion Specifications[11] of the Industrial Digital Twin Association (IDTA),[12] one could address interoperability issues within and among platforms.

Industrial platforms also pay more attention to data collection, storage or analysis (data processing overview T11.a), highlighting the aim of industrial IIoT platforms to cover the entire data lifecycle. However, seamless real-time processing from edge to cloud is still a challenge [6]. As illustrated in Fig. 8, it is notable that the rate of use of data collection, storage and analysis in scientific platforms correlates with the rate of use of data collection in the included industrial platforms. Use and adoption of AI (T12) is a closely related topic, with more industrial platforms offering abilities here (such as anomaly detection, behavior prediction, interfaces to AI services), while AI became a must-have topic. However, we are missing re-usable or easy-to-use AI (one notable exception is Amazon IoT) and real-time support. The usage of external AI applications is notably higher in scientific platforms.

Regarding openness and extensibility (T13), we found that industrial platforms significantly provide better support. Industrial platforms not only prioritize the needs of developers but also encourage innovation and customization. The substantial difference in configurability (T14) percentages indicates that industrial platforms offer more extensive options for customization. This flexibility is vital for accommodating the varied and often complex requirements of industrial processes. However, more details on the employed techniques as well as more

[11] https://opcfoundation.org/about/opc-technologies/opc-ua/ua-companion-specifications/.

[12] https://industrialdigitaltwin.org/.

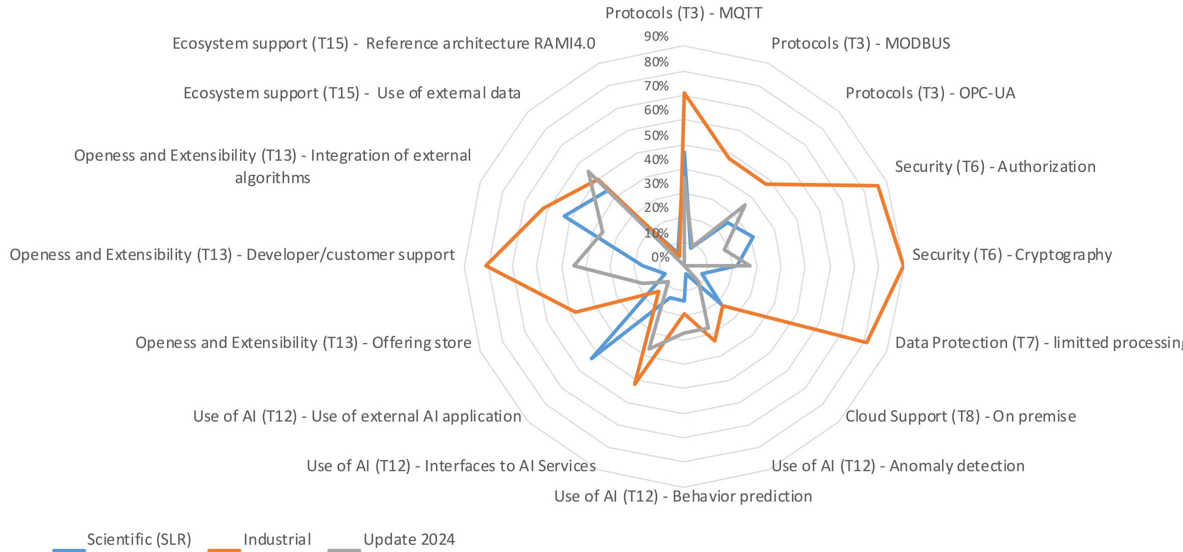

Fig. 9. Comparison of both sides based on answers to analyzed sub-topics. The radar chart further contains the results of the SLR update (Section 5.4) where scientific papers published in 2024 are analyzed and reported.

intensive work on automated configurability ensuring consistency even in (interconnected) ecosystems are desirable.

The support for ecosystems (T15) is a critical factor in the collaborative and interconnected landscape of IIoT. While both sides demonstrate a commitment to fostering ecosystems, the industrial side suggests more dedication. Here, too, more research and standardization is needed.

Another dimension for comparing the scientific and the industrial side is their development over time and mutual influences. As our summary of the scientific approaches in Section 5.1.6 identified research topics per publication year, we can compare the scientific aims since 2019 with our industrial findings in [7] from 2021. While industrial platforms take up individual open source components, complete open source platforms, open license models or truly interoperable open platform ecosystems are still missing, although desired by the users [6]. Also topics like blockchains or model-driven development are not or only rarely taken up. While most industrial platforms claimed AI or edge(-cloud) computing as important trends, only a minority realized (user-defined) AI on the edge. Further, the trend of digital twins, in particular AAS, is recently taken up in scientific approaches. This was already anticipated by some industrial platforms, e.g., by concepts like “things modeling”, while an interoperable standardization of digital twins and an uptake in industrial platforms, e.g., in terms of the AAS, is still missing.

Besides our analysis topics, comparisons of scientific and/or commercial platforms for further topics would be desirable, among them practical comparisons regarding costs, flexibility, runtime characteristics, AI or manufacturing aspects. However, an own analysis or experimentation with individual platforms is out of the scope of this literature-based article.

7. Threats to validity

Systematic literature reviews are subject to a typical set of threats to validity, e.g., as surveyed in [80]. We mainly focus on the SLR threats in the dimensions of construct, internal, conclusion and external validity and briefly summarize relevant aspects for the industrial comparison in Section 7.5.

7.1. Construct validity

As the *source repositories may be incomprehensible*, we used three renowned repositories, on which we conducted a pre-study achieving 100% coverage with expected papers. Further, the actual coverage is larger as the repositories are cross-linked. However, as discussed in Section 4.1, there is a topic-induced bias for IEEE. As for any SLR, we had to trade off time, effort and information value: We applied usual criteria, e.g., to exclude gray literature, non-English work or short papers. While lower quality of gray literature may affect our results adversely, short papers, as we experienced, typically do not contain enough details for our analysis. To prevent *inappropriate or incomplete search terms*, we derived them systematically and assessed them in our pre-study. Regarding further search strings, we made a similar trade-off for the repositories. As a mitigation, we applied backward snowballing, which added only 5 included papers, indicating a convergence of the results. Further, *inappropriate analysis topics, exclusion or inclusion criteria* may lead to bias and wrong filtering or review decisions. In our case, the analysis topics stem from a vision discussion between industry and research, and, thus, are not arbitrary. However, still relevant topics may be missing, while the open collection of further abilities (T16) did not reveal obvious gaps. Also exclusion or inclusion criteria may have inadequately affected the results. For this purpose, we intentionally applied only a small set of usual criteria.

7.2. Internal validity

All included papers report on positive results, none about failures and, thus, may be subject to *publication bias*. We consider this threat to be low, as we aimed for determining the capabilities of realized/existing approaches. Further, a *subjective or biased data extraction* may lead to wrong results. To prevent this, we jointly prepared an extensive review guideline and a comprehensive spreadsheet to systematically collect data on all SLR filtering steps. All recorded information was validated by a second author and discrepancies were discussed and resolved. We also applied an early validation after processing 10 papers to prevent misunderstandings. To fight *subjective quality assessment*, we defined measurable quality criteria (QA1–QA6) and applied a similar validation and agreement procedure.

7.3. Conclusion validity

For most analysis topics, free text or lists of characteristics had to be extracted. Only for a few topics, we prescribed an initial classification based on [7] and allowed for re-classification during data extraction or analysis. To avoid *misclassification of information*, we validated the results by mutual agreements. *Primary study duplication* may occur as the bibliographic data in the paper repositories is sometimes incomplete, incorrect or follows different conventions. We analyzed all bibliographic data by tool support for duplicates and corrected information or removed duplicates as early as possible. Also *subjective interpretation of data* may cause this threat. We applied formalized quality rules (cf. Section 4.3), cross-checking and joint decision making for disagreements on all data as mitigation.

7.4. External validity

The selected publication *time span* may have been too short, i.e., missing information may affect generalizability: We based the time span on the first occurrence of "Industry 4.0" [5] and the distribution of publication years in Fig. 2 indicates that we covered the relevant time. Further, the information in *unavailable papers* may contribute to this threat. As discussed in Section 4.3, only 4% of all papers from the first stage were not available to us, a share that we already minimized through author contacts or community portals. *Primary generalizability* may be affected as an SLR is always based on a timed snapshot. To counter this, we would have to analyze as many papers as possible, which contradicts the necessary trade-off of time, effort and value. We believe that the presented information allows characterizing the field at the time of publication. However, the research field is evolving and so there may be future work with better coverage of our analysis topics or new/different properties that we did not cover.

7.5. Industrial platforms

For the comparison with industrial work, a bias may be the focus on platforms with the largest market share or the relevancy to IIP-Ecosphere. The underlying material, which was mostly technical, not peer-reviewed and obtained already in 2020, was reviewed more informally without a cross-validation. However, we aimed at characterizing industrial platforms based on vendor statements in a rapidly evolving market [30] and, in this context, addressed the most relevant threats [7].

8. Conclusion

IIoT, Industry 4.0 or CPPS software platforms are one core ingredient for smart manufacturing systems. While industrial platforms are often bound to business models and, thus, may focus on their own devices or proprietary protocols, scientific approaches can act more freely and may integrate various approaches to showcase novel capabilities. Consequently, there are significant differences between the research and industrial platform approaches, but these sides also share trends, concepts and abilities.

In this article, we reported on a systematic literature review of the research publications of eight years, starting with 1620 candidates that we finally filtered down to 36 papers. We complemented the SLR by a SLR update through a forward-snowballing of the 36 papers, leading to more than 330 candidates, from which we identified further 11 relevant papers. We analyzed these papers along 16 topics, providing an overview of the research landscape (RQ1), the context the work is currently happening within (RQ2) as well as an analysis of actual capabilities and differences (RQ3). Further, we compared the results of 13 analysis topics with 21 industrial platforms and discussed from our point of view commonalities, differences, trends as well as potential future topics (RQ4). In particular, we miss more open source platform efforts, the uptake of novel, more interoperable protocols and information models, open use of devices from AI on the edge toward a shared edge–cloud-continuum, end-to-end use of data protection and data sharing methods, interoperable and standardized use of digital twins, e.g., for integrated simulations and pre-deployment testing, re-usable and easy-to-use AI toolkits, effective approaches on configurability and reduction of development efforts (e.g., low/no code) as well as more virtual interaction and integration of ecosystems or data spaces. Further, we miss systematic evaluations of realized platform capabilities including runtime characteristics and, as a prerequisite for replicated evaluations, increased publication of implementation and evaluation artifacts.

In future work, we plan to take the insights from this discussion of the IIoT platform landscape into the design and realization of our open source AI-enabled Industry 4.0 platform oktoflow [12] and strengthen the aspect of the edge–cloud-continuum as well as (explainable) AI and MLOps. We can also imagine, besides runtime evaluations of our own oktoflow components like [81], experimental comparisons with other platforms. Moreover, we aim to focus more on security and privacy aspects. The Network and Information Security (NIS) Directive mandates EU Member States to adopt and publish proper measures. We will investigate the effect of this directive on IIoT platforms. Further, it is interesting to investigate, support and adopt efforts and guidelines by the Industry 4.0 community, e.g., on IDTA formats around Asset Administration Shells, the unified data space governance model by Data Space 4.0 or the joint use of data in production by Manufacturing-X.

CRediT authorship contribution statement

Holger Eichelberger: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Christian Sauer:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation, Conceptualization. **Amir Shayan Ahmadian:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation, Conceptualization. **Christian Kröher:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The collected SLR data is available via Zenodo.

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